

Decisions, decisions: Achieving near perfect page prefetching using decision trees

Abstract

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1 Introduction

Increasing memory requirements for applications, e.g., machine learning, coupled with the slowdown of DRAM scaling [12, 13], makes DRAM one of the costliest components in data centers, constituting as much as 30% of the entire cost [20]. Thus prefetching is essential to reduce the number of page faults and improve application performance.

NEED TO ADD MORE CONTEXT HERE to explain why we need prefetching for page faults.

Prefetching is a very rich field with many different approaches. Traditional prefetching algorithms use heuristics to predict the next page to prefetch. These heuristics are based on the recent access patterns of the application and guess the next page to prefetch based on these patterns using a pre-defined algorithm [2, 9].

With the recent bloom of machine learning (ML) techniques, researchers have started to apply ML to the problem of prefetching.

WHAT ARE WE ACTUALLY PROPOSING?

In this paper, we explore the use of machine learning to predict page accesses and prefetch pages with high accuracy. In particular, compare LSTM, Transformer or Large Language Models for prefetching.

2 Background and Related work

2.1 Heuristic Page prefetching

Page prefetchers are used in most modern operating systems to reduce the latency of page access from swap. Traditional algorithms prefetch based on sequential access to virtual addresses [5, 10] and are successful at fetching spatially related

pages. One of the recent state-of-art prefetcher, Leap [1], improved traditional prefetching using majority trend detection to identify stride patterns; this makes Leap resilient to short-term irregularities in the memory access stream. Leap improved performance for many applications but cannot prefetch irregular accesses. Multiple researchers have tried to improve Leap's performance by addressing its shortcomings [18, 19, 22] in

- prefetching irregular patterns,
- isolating the swap subsystem and memory access histories of threads, and
- coordinating memory accesses from the garbage collector and the application.

All the above resulted in the non-perfect prefetching performance of the prefetcher in our initial experiments.

HOW TO CONTRAST OUR WORK WITH TRADITIONAL PREFETCHING?

Studies [15] has shown the need for more aggressive prefetching to reduce the number of page faults while sacrificing some portability and memory bandwidth. Thus, we propose a machine learning based prefetcher that can improve the prefetching accuracy and reduce the number of page faults.

2.2 Machine Learning Prefetching

Past success of machine learning in cache eviction [17] and cache prefetching [7] has shown that machine learning can be used to predict page accesses patterns with high accuracy.

NEED MORE PREVIOUS WORKS

2.2.1 LSTM based prefetching. LSTM, first introduced in 1997 [8], is a type of recurrent neural network that is capable of learning long-term dependencies. An LSTM is composed of a cell c , a hidden state h , an input gate i , an output gate o , and a forget gate f . The cell remembers values over arbitrary time intervals and the three gates regulate the flow of information into and out of the cell. The hidden state h is recurrent and is passed to the next time step. The process of an LSTM to process an input x_t at time step t is as follows:

1. Compute the input gate i_t , forget gate f_t , and output gate o_t in parallel.

$$i_t = \sigma(W_{xi}x_t + W_{hi}h_{t-1} + W_{ci}c_{t-1} + b_i)$$

$$f_t = \sigma(W_{xf}x_t + W_{hf}h_{t-1} + W_{cf}c_{t-1} + b_f)$$

$$o_t = \sigma(W_{xo}x_t + W_{ho}h_{t-1} + W_{co}c_{t-1} + b_o)$$

Here, W and b are the weights and biases of the gates, and σ is the sigmoid function.

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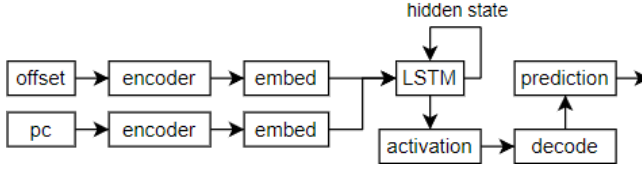


Figure 1. LSTM based prefetcher architecture.

2. Update the cell state c_t and hidden state h_t .

$$c_t = f_t \odot c_{t-1} + i_t \odot \tanh(W_{xc}x_t + W_{hc}h_{t-1} + b_c)$$

$$h_t = o_t \odot \tanh(c_t)$$

Here, $\odot$ is the element-wise multiplication operator.

LSTM has shown promising results in cache prefetching [7] due to its ability to learn long-term dependencies. However, LSTM has a large number of parameters and is computationally expensive to train and run.

2.2.2 Transformer based prefetching. CHECK OUT THIS PAPER [6] TO BE ADDED

2.2.3 Large Language Models based prefetching. TO BE ADDED

3 Problem Statement

WHAT MOTIVATES OUR WORK?

Possible accessible virtual memory pages for a program is huge. It is normal for program to access several 10th of GBs of memory. Then that would be easy to get a billion of possible pages the program will access. It is hard to predict which page the program will access next and to prefetch it before the program access it.

3.1 Information to be used for prefetching

WHAT INFORMATION CAN BE USED FOR PREFETCHING?

3.2 Prefetching as a classification problem

Past work [7] suggested that although page addresses are number, regression models are not suitable for prefetching. Instead, prefetching can be treated as a classification problem.

We treat the address space as a large, discrete vocabulary and perform classification.

4 Design and Implementation

In this section, we introduce the design and implementation of our LSTM, Transformer, and LLM based prefetchers.

4.1 LSTM based Prefetcher

We implement an LSTM based prefetcher based on past work [7]. It is shown at a high level in Fig. 1.

Each input offset, output offset and pc are encoded as a one hot representation of the total classes. Then the one

hot encoding is embedded in a high dimensional space. The embeddings are then concatenated and fed to the LSTM. The LSTM outputs the probability of the next page to prefetch. To get multiple pages to prefetch, we can apply softmax on the output and sample from the distribution.

- Embedding dimension: 128
- LSTM hidden dimension: 128
- Number of LSTM layers: 2

5 Evaluation

Correct prefetching choices can significantly reduce the number of page faults and improve the performance of applications. However, wrong prefetching choices can lead to unnecessary page faults due to memory pollution and degrade performance. In this section, we evaluate the performance of the proposed ML prefetchers on a set of workloads and compare them with the state-of-the-art prefetcher, LEAP [1].

5.1 Metrics

We use memory access traces of benchmark workloads to evaluate the performance of the proposed ML prefetchers. The metrics we will focus on is:

Definition 5.1. Coverage = $\frac{\text{Page faults predicted by Prefetching}}{\text{Total Page faults}}$

Definition 5.2. Accuracy = $\frac{\text{Page faults predicted by prefetching}}{\text{Total predictions}}$

Coverage measures the percentage of page faults that were satisfied by the prefetcher. A higher coverage indicates that the prefetcher is able to predict more page faults. Accuracy measures the percentage of correct predictions made by the prefetcher. A higher accuracy indicates that the prefetcher is making correct predictions.

We want both coverage and accuracy to be high. However, there is a trade-off between the two. A prefetcher can achieve high coverage by prefetching aggressively, but this may lead to a decrease in accuracy. On the other hand, a prefetcher can achieve high accuracy by prefetching conservatively, but this may lead to a decrease in coverage.

We focus on the coverage metric as it is more important for prefetchers in our experiment since none of our prefetchers are aggressive. **TODO: HOW TO WORD THIS BETTER?**

5.2 Data Collection and processing

To collect page faults, we use fltrace [21], which will interpose on all **heap allocations**. Thus, we can collect page faults for all heap accesses. Stack accesses are excluded from the analysis since stack page faults are rare and are not the focus of this work.

Pervious work [7] has suggested that offsets of page address accesses works better than absolute addresses, and since we are doing classification, we need to filter out the rare classes and only leave up to 50,000 most occurred classes for offsets and pc values.

Definition 5.3. Offset = Next Page Address - Last Page address

fltrace will need local RAM size to be set to run. It simulates the physical memory the program can access. Since each workload has a different memory requirement, we set the local RAM size to be 25% and 50% of the maximum memory requirement of the workload. This can be found by running `/usr/bin/time -v ./workload` and looking at the Maximum resident set size.

5.3 Workloads

We chose the following workloads for our evaluation:

- SPEC2017 [4]: mcf omnetpp and lbm with the default input files offered by the benchmark suite.
- GAP [3]: the default bfs, pr, bc workloads on the twitter dataset [11].
- WiredTiger [14]: the default ycsb-a and ycsb-c workloads with `icount=30000000` in the benchmark configuration file.
- DiLos-redis [16, 22]: the redis-benchmark workload with `lrange` on a list of queries made by DiLOS [22].

5.4 Analysis

TODO: Wait for all my results to come in.

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